Predicting Insurance Premiums

Crystal Chang, Lauren Le, Egnegt Odbayar, Ada Phoo,
Samantha DeMars, & Rachie Yim

Overview:

01

Introduction

02

Data Preprocessing

03

EDA + Multicollinearity

04

F-test & R-adjusted Values

05

Gender + Cross Validation

06

Unseen Data Results

Introduction

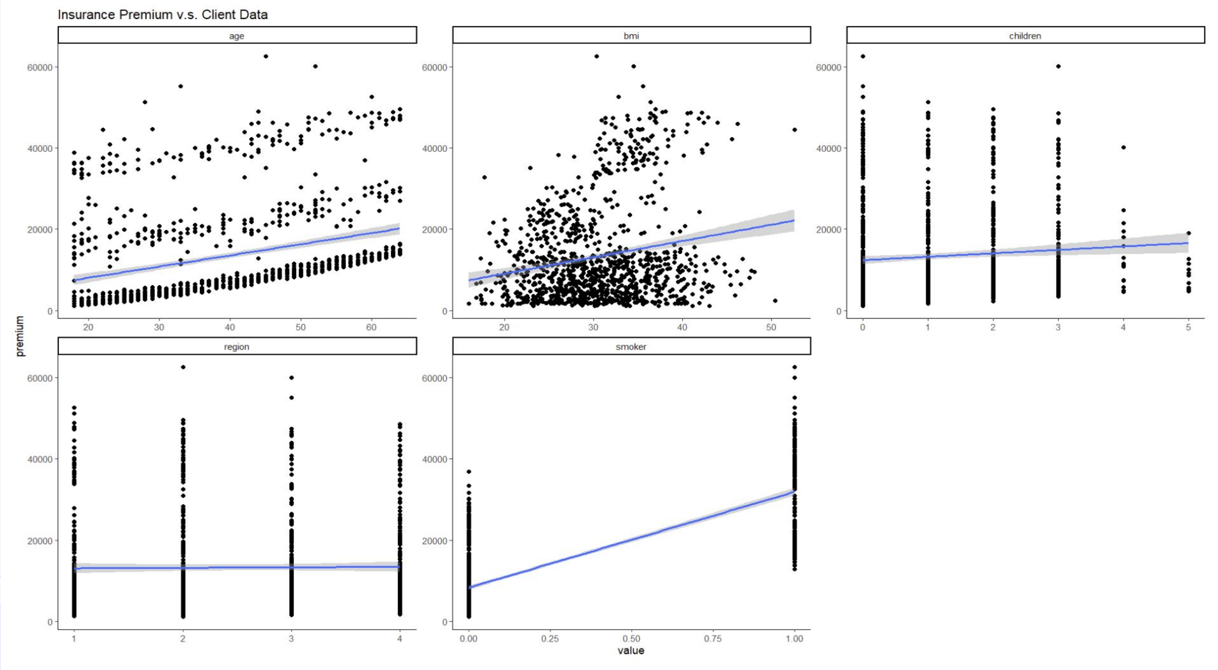
Health insurance premiums can vary widely based on a multitude of factors. The healthcare dataset we will be analyzing includes insurance charge data from patients of different age, sex (male, female), BMI, smoking status, number of children, and region of residence (southwest, southeast, northwest, northeast).
We will build a machine learning model to predict insurance charges and assess the relationships between these variables and the medical costs. For the gender factor, we will employ cross-validation techniques.

We will build a model and perform an F-test using a 0.05 significance level. To evaluate the model, we use two key metrics: R-squared (R^2) and the Root Mean Squared Error (RMSE). This comparison of cross-validation helps us assess the generalizability of the model for noticing trends in gender. Based on our analysis, we can refine our final model to determine which factors most significantly impact individual health insurance costs, and possibly predict insurance changes based on demographic and health related variables.

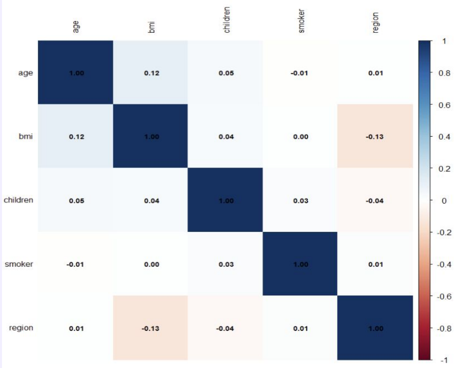
Data – Preprocessing

- Encoding categorical data
  - Converted categorical variable **smoker** into binary data to be able to run our analysis
    - smokers were assigned a value of 1 and non-smokers a value of 0
  - Assigned each value of **region** to a corresponding numerical value
    - southwest region was assigned a value of 1, southeast a value of 2 , etc
- Removed **sex** as a variable to be able to analyze people as a collective
- Splitting the data into testing set and training set
  - Ensured sample was random
  - Split 20% for testing, 80% for training

EDA + Multicollinearity



Correlation:
Smoker: 0.79
Age: 0.32
BMI: 0.20
Children: 0.08
Region: 0.01



F-test & R-adjusted Values

Full Model Adjusted R^2: 0.7574098

F test: full model vs reduced model 1 (no region)

P-value: 0.1648 F stat: 1.9322

Reduced Model 1 Adjusted R^2: 0.7571975

F test: full model vs reduced model 2 (no region, no children)

P-value: 0.01031 F stat: 4.5948

Reduced Model 2 Adjusted R^2: 0.7557737

Gender & Cross Validation

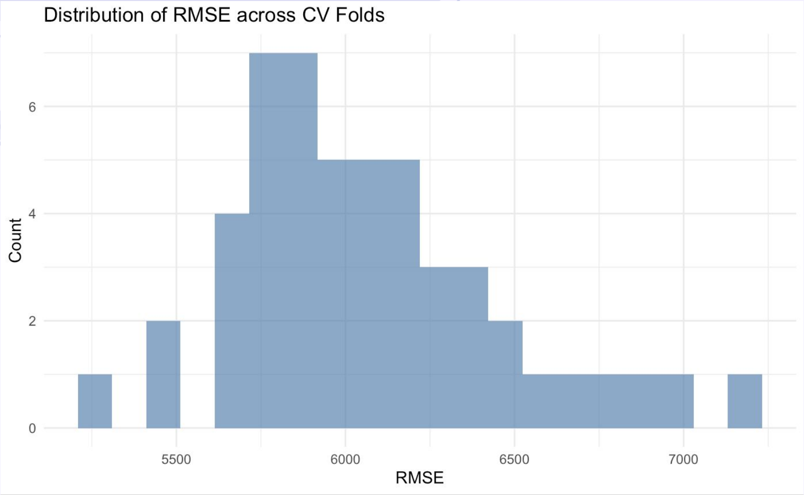
Question 3: Does Gender play a role in an individual's medical costs?

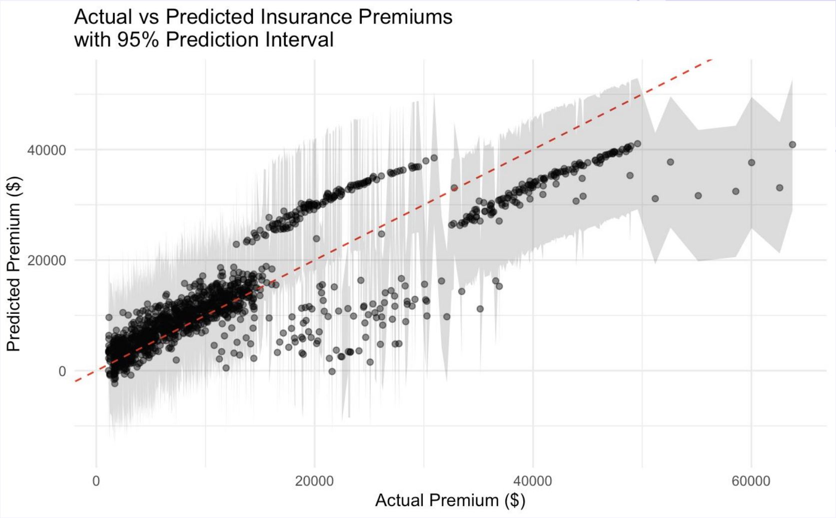
- F-Test (model with gender):
  - P-value: 0.7684

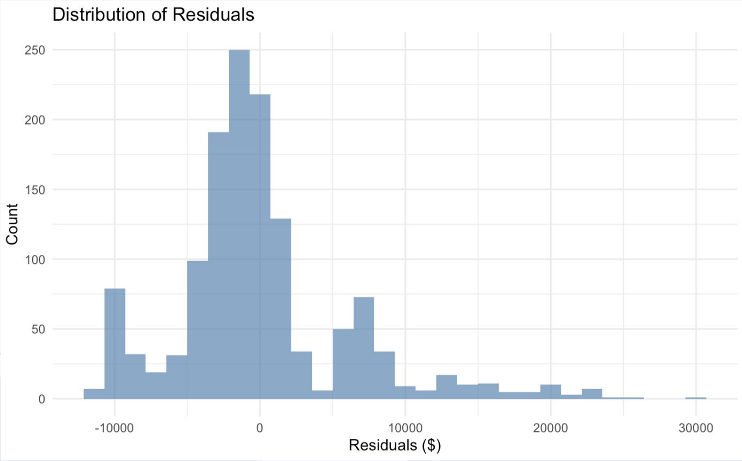
Our testing model/cross-validation results vs Unseen data results

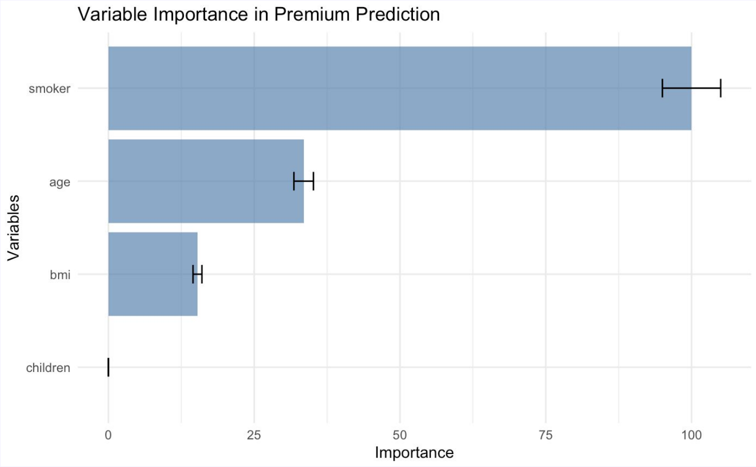
- CV results:
  - R squared: 0.75
  - RMSE(*Root Mean Squared Error*): $6,067
  - MAE(*Mean Absolute Error*): $4,193
- Unseen data results:
  - R squared: 0.72
  - RMSE: $6,354
  - MAE: $4,321

Going back to our first question – Can we predict insurance charges based on demographic and health-related variables?


Distribution of RMSE across CV Folds
Count
0
2
4
6
5500
6000
6500
7000
RMSE


Actual vs Predicted Insurance Premiums
with 95% Prediction Interval
Predicted Premium ($)
0
20000
40000
0
20000
40000
60000
Actual Premium ($)


Distribution of Residuals
Count
0
50
100
150
200
250
-10000
0
10000
20000
30000
Residuals ($)


Variable Importance in Premium Prediction
Variables
smoker
age
bmi
children
0
25
50
75
100
Importance